

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

1

A curve is defined implicitly. Find dy/dx in terms of x and y .

[5 marks]

$$x^2 + xy + y^2 = 7$$

2

Find the area enclosed between the two curves.

[6 marks]

$$y = x^2, \quad y = 2x$$

3

Evaluate the limit and justify any algebraic simplification.

[4 marks]

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$

4

Differentiate the function and simplify your answer.

[5 marks]

$$y = (3x^2 + 1)^5$$

5

A rectangle has perimeter 40 cm. Find the dimensions that maximise its area, and show your result is a maximum.

[6 marks]

6

Evaluate the definite integral, showing the antiderivative you use.

[5 marks]

$$\int_0^2 (3x^2 - 4x + 1) dx$$

7

Find the equation of the tangent to the curve below at the point where $x = 2$.

[5 marks]

$$y = x^3 - 2x$$

8

Solve the differential equation given that $y(0)=2$.

[6 marks]

$$\frac{dy}{dx} = 3y$$
